Formalizing the Exponential Blowup in the Transformations between CNF and DNF

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Abstract

A well-known folklore result about propositional logic is that transforming a formula in disjunctive or conjunctive normal form can lead to an exponential blowup of the formula size. We formalize this result in the form of two theorems in Isabelle/HOL and discuss the challenges we encountered.

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1 Introduction

The metatheory of propositional logic has been studied for many decades and has been the subject of extensive formalization effort for many years. However, to the best of our knowledge, very few quantitative results have been formalized. One well-known quantitative folklore result is that transforming a formula in disjunctive or conjunctive normal form can lead to an exponential blowup of the formula size.

In this paper, we first briefly cover the necessary mathematical background (Section 2) and then report on our Isabelle/HOL formalization of this exponential blowup. We start by presenting a few preliminary definitions and lemmas (Section 3).

We then prove the exponential blowup when converting a formula from conjunctive to disjunctive normal form (Section 4). We do this in two steps: First, we prove the low-level Proposition 17 that assumes the existence of two equivalent formulas fulfilling some assumptions and that exhibits the exponential lower bound on the formula size. Second, we use this proposition to prove in the desired Theorem 18 that such formulas always exist.

We analogously prove the same exponential blowup when converting a formula from disjunctive to conjunctive normal form (Section 5). We do this again in two steps: First, the low-level Proposition 24 assumes again the existence of two equivalent formulas fulfilling some assumptions and exhibits the exponential lower bound on the formula size; the proof uses Proposition 17. Second, we use Proposition 24 to prove in the desired Theorem 25 that such formulas always exist.

Finally, we discuss (Section 6) some of the challenges we encountered when proving statements about the size of formulas.

Our work is part of the IsaFoL (Isabelle Formalization of Logic) effort [1]. Our formalization consists of ???? nonempty lines[1] and is available online [10]; we will soon submit it to the *Archive of Formal Proofs*. It was developed with Isabelle2025 [6, 7] and with extensive usage of Sledgehammer [2, 8] and the structured Isar language [11]. In this paper, we only provide some interesting lemmas and proof sketches; we refer interested readers to the formalization for more details.

2 Background

A *formula* of propositional logic is defined inductively as either falsehood ($\bot$), an atom (or *variable*), the negation ($\neg$) of a formula, or the disjunction ($\vee$), conjunction ($\wedge$) or implication ($\rightarrow$) of two formulas. A *literal* is either an atom x, a negated atom $\neg x$, falsehood $\bot$, or negated falsehood $\neg\bot$. The literals x and $\neg x$ as well as $\bot$ and $\neg\bot$ are *complements* of each other.

A *valuation* is a function from atoms to Boolean values. The *evaluation* of a formula w.r.t. a valuation is the Boolean value obtained after replacing all atoms with the specific truth values assigned by the valuation and evaluating according to the usual Boolean evaluation rules.

We say that a valuation α *satisfies* a formula ϕ, written $\alpha \models \phi$, if the formula evaluates to true w.r.t. α. A formula ϕ is *satisfiable* if there exists a valuation that satisfies it (i.e., $\exists\alpha.\ \alpha \models \phi$) and *tautological* if it is satisfied by every valuation (i.e., $\forall\alpha.\ \alpha \models \phi$). Two formulas ϕ and ψ are *equivalent* if they evaluate to the same truth value for every valuation (i.e., $\forall\alpha.\ \alpha \models \phi \longleftrightarrow \alpha \models \psi$).

A formula ϕ is in *negation normal form* if ϕ contains no implications and if negations only occur immediately in front of atoms or falsehood (i.e., ϕ is built from literals by conjunction and disjunction only).

For a formula ϕ in negation normal form we define the dual formula $\bar{\phi}$ in negation normal form by replacing every literal by its complement and interchanging the symbols $\wedge$ and $\vee$. For every formula ϕ in negation normal form, $\bar{\phi}$ is equivalent to $\neg\phi$.

▶ **Definition 1.** We define the following classes of formulas:
- A *term* is a conjunction $a_1 \wedge \ldots \wedge a_k$ of literals.
- A *clause* is a disjunction $a_1 \vee \ldots \vee a_k$ of literals.
- A formula in *disjunctive normal form* (DNF) is a disjunction $T_1 \vee \ldots \vee T_m$ of terms.
- A formula in *conjunctive normal form* (CNF) is a conjunction $C_1 \wedge \ldots \wedge C_m$ of clauses.

Note that every formula in disjunctive or conjunctive normal form is also in negation normal form. For a formula ϕ in conjunctive normal form, the dual formula $\bar{\phi}$ is in disjunctive normal form, and vice versa.

It is well known that every formula of propositional logic can be transformed into an equivalent formula in disjunctive normal form and into an equivalent formula in conjunctive normal form. However, the cost of these transformations is an exponential increase in the formula size in the worst case (see, e.g., the textbook of Kleine Büning and Lettmann [3]).

▶ **Theorem 2.** *For all sufficiently large $n \in \mathbb{N}$, there are formulas F_n in conjunctive normal form with $|F_n| = O(n)$, such that every equivalent formula G_n in disjunctive normal form is of size $|G_n| \geq n \cdot 2^n$.*

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[1] Counted using `grep -Ev '^[[:blank:]]*$'`.

Proof. The formulas F_n are defined as $F_n = \bigwedge_{i=1}^{n}(x_{i,0} \vee x_{i,1}) \wedge (\neg x_{i,0} \vee \neg x_{i,1})$.

If $G_n = \bigvee_{j \le m} T_j$ is a formula in conjunctive normal form equivalent to F_n, then the following two claims hold, which together prove the statement.

Claim 1. Each of the terms T_j contains exactly one of the literals $x_{i,0}$ and $x_{i,1}$, for every $i \le n$; otherwise, there is a valuation that satisfies this term, and thus G_n, but does not satisfy F_n. It follows that every term T_j is of size at least n.

Claim 2. For every vector $\epsilon = (\epsilon_1, \ldots, \epsilon_n) \in \{0,1\}^n$, there is one of the terms T_j in G_n, which contains the literals $x_{1,\epsilon_1}, \ldots, x_{n,\epsilon_n}$; otherwise, we can find a valuation that satisfies F_n, but does not satisfy G_n. It follows that G_n has at least $m \ge 2^n$ many terms. ◀

▶ **Theorem 3.** *For all sufficiently large $n \in \mathbb{N}$, there are formulas F'_n in disjunctive normal form with $|F'_n| = O(n)$, such that every equivalent formula G_n in conjunctive normal form is of size $|G_n| \ge n \cdot 2^n$.*

Proof. Let $F'_n = \bar{F}_n$ for the formula F_n from the proof of Theorem 2. If there was a formula G_n in conjunctive normal form equivalent to F'_n with $|G_n| < n \cdot 2^n$, then $\bar{G}_n$ would be a formula in disjunctive normal form equivalent to F_n with $|\bar{G}_n| = |G_n| < n \cdot 2^n$, contradicting Theorem 2. ◀

3 Preliminaries

We reused the theories of proposition formulas from Michaelis and Nipkow's formalization of proof systems for classical propositional logic [4, 5]. Formulas have type *'a formula*, where *'a* is an abstract type of atoms. Basic operations on formulas and related lemmas are expressed for any atom type. However, our main theorems instantiate the atom type with variables $x_{i,b}$, where i is a natural number and b is a Boolean value.

A formula is *conjunctive* if it is either a literal or its first subformula is a literal and its second a conjunctive formula. Analogously a formula is *disjunctive* if it is either a literal or its first subformula is a literal and its second a disjunctive formula.

We now define some operations to build the conjunction and disjunction of a list of formulas.

▶ **Definition 4.** The functions $\mathsf{BigAnd}' :: \mathit{'a\ formula\ list} \Rightarrow \mathit{'a\ formula}$ and $\mathsf{BigOr}' :: \mathit{'a\ formula\ list} \Rightarrow \mathit{'a\ formula}$ respectively build the conjunction and disjunction of all formulas in a list:[2]

$$\begin{aligned} \mathsf{BigAnd}'\ [] &= \neg\bot & \mathsf{BigOr}'\ [] &= \bot \\ \mathsf{BigAnd}'\ [\phi] &= \phi & \mathsf{BigOr}'\ [\phi] &= \phi \\ \mathsf{BigAnd}'\ (\phi \,\#\, \phi s) &= \phi \wedge \mathsf{BigAnd}'\ \phi s & \mathsf{BigOr}'\ (\phi \,\#\, \phi s) &= \phi \vee \mathsf{BigOr}'\ \phi s \end{aligned}$$

In the rest of this paper, we write $\bigwedge_{i=0}^{|\phi s|} \phi s_{[i]}$ for $\mathsf{BigAnd}'\ \phi s$ and $\bigvee_{i=0}^{|\phi s|} \phi s_{[i]}$ for $\mathsf{BigOr}'\ \phi s$.

▶ Remark. Our formalization always uses nonempty lists and the base cases of Definition 4 could be be left unspecified. However, providing these base cases allows us to prove many lemmas for arbitrary lists without requiring a nonemptyness assumption that would need to be repeatedly discharged.

We will later need to generate lists of formulas for BigAnd′ and BigOr′ and define two functions to do so.

[2] The infix operation $x \,\#\, xs$ adds an element x at the front of a list xs.

▶ **Definition 5.** The functions unconj :: *'a formula* ⇒ *'a formula list* and undisj :: *'a formula* ⇒ *'a formula list* respectively extract the list of subformulas directly under the conjunctions and disjunctions of a given formula:[3]

$$\mathsf{unconj}\ (\phi \wedge \psi) = \mathsf{unconj}\ \phi \mathbin{@} \mathsf{unconj}\ \psi \qquad \mathsf{undisj}\ (\phi \vee \psi) = \mathsf{undisj}\ \phi \mathbin{@} \mathsf{undisj}\ \psi$$
$$\mathsf{unconj}\ \phi = [\phi] \qquad \mathsf{undisj}\ \phi = [\phi]$$

We will also need to construct formulas of a very specific family and define a function to do so.

▶ **Definition 6.** The function Fn :: *nat* ⇒ *'a formula* creates a formula containing the conjunction of the exclusive disjunctions of each variable pair $x_{i,false}$ and $x_{i,true}$ with $i \in \{1, \ldots, n\}$:

$$\mathsf{Fn}\ 0 = \neg\bot$$
$$\mathsf{Fn}\ (n+1) = ((x_{i,false} \vee x_{i,true}) \wedge (\neg x_{i,false} \vee \neg x_{i,true})) \wedge \mathsf{Fn}\ n$$

In the rest of this paper, we write F_n for $\mathsf{Fn}\ n$.

Definition 6 is chosen such that a formula F_n is in conjunctive normal form and can only be satisfied by valuations that assign different truth values to the variables $x_{i,false}$ and $x_{i,true}$.

▶ **Lemma 7.** *Let n be a natural number. The formula F_n is in conjunctive normal form.*

▶ **Lemma 8.** *Let n be a natural number. Let α be a valuation. The valuation α satisfies F_n if and only if $(\alpha\, x_{i,false}) \neq (\alpha\, x_{i,true})$ for every $1 \leq i \leq n$.*

Finally, we provide two different definitions for the size of a formula.

▶ **Definition 9.** The function size :: *'a formula* ⇒ *nat* counts the number of atoms, ⊥, negations, disjunctions, conjunctions, and implications in a formula. In the rest of this paper, we write $|\phi|$ for $\mathsf{size}\ \phi$.

▶ **Definition 10.** The function sizef :: *'a formula* ⇒ *nat* counts the number of atoms, ⊥, disjunctions, conjunctions, and implications in a formula; the negations are ignored.

Because sizef ignores negations, it always evaluates to a smaller number than $|\cdot|$.

▶ **Lemma 11.** *Let ϕ be a formula.* $\mathsf{sizef}\ \phi \leq |\phi|$.

We can observe the difference and similarity between the two definitions by looking at the sizes of the F_n formula and the conjunction and disjunction of lists of formulas.

▶ **Lemma 12.** *Let n be a natural number.* $\mathsf{sizef}\ F_n = 8 \cdot n + 1$ *and* $|F_n| = 10 \cdot n + 2$.

▶ **Lemma 13.** *Let ϕ be a formula in conjunctive normal form and ψ be a formula in disjunctive normal form. The following equalities hold:*

$$\mathsf{sizef}\ \left(\bigwedge\nolimits_{i=0}^{|\mathsf{unconj}\ \phi|} (\mathsf{unconj}\ \phi)_{[i]}\right) = \mathsf{sizef}\ \phi \qquad \left|\bigwedge\nolimits_{i=0}^{|\mathsf{unconj}\ \phi|} (\mathsf{unconj}\ \phi)_{[i]}\right| = |\phi|$$
$$\mathsf{sizef}\ \left(\bigvee\nolimits_{i=0}^{|\mathsf{undisj}\ \psi|} (\mathsf{undisj}\ \psi)_{[i]}\right) = \mathsf{sizef}\ \psi \qquad \left|\bigvee\nolimits_{i=0}^{|\mathsf{undisj}\ \psi|} (\mathsf{undisj}\ \psi)_{[i]}\right| = |\psi|$$

Most of the statements about the size of formulas are expressed in terms of sizef but, luckily, Lemma 11 will allow us to remove sizef entirely from the statements of Theorems 18 and 25.

[3] The infix operation $xs \mathbin{@} ys$ appends a list xs at the front of a list ys.

4 CNF to DNF (MDS)

Our formalization splits the paper proof of Theorem 2 in a low-level proposition, which assumes the existence of formulas with specific shapes and properties, and a theorem, which proves that such formulas can always be obtained. We start by proving a few basic properties of conjunctive formulas w.r.t. valuations.

We first prove a monotonicity property of satisfiability w.r.t. valuations.

▶ **Lemma 14.** *Let ϕ be a conjunctive formula. Let α and β be valuations. If both valuations map every variable occurring in ϕ to the same Boolean value and if α satisfies ϕ, then β also satisfies ϕ.*

Next, we prove some sufficient conditions for a valuation to not satisfy a conjunctive formula.

▶ **Lemma 15.** *Let ϕ be a conjunctive formula. Let α be a valuation. If*

- *there exists a variable that occurs positively in ϕ and is mapped to false by α or*
- *there exists a variable that occurs negatively in ϕ and is mapped to true by α,*

then α does not satisfy ϕ.

Next, we prove a lower bound of the size of $\bigvee_{i=0}^{|\phi s|} \phi s_{[i]}$.

▶ **Lemma 16.** *Let $n > 0$ be a natural number. Let ϕs be a list of formulas. If every $\phi \in \mathsf{set}\, \phi s$ has $\mathsf{sizef}\ \phi \geq m$ and if $|\phi s| \geq 2^n$, then $\mathsf{sizef}\ (\bigvee_{i=0}^{|\phi s|} \phi s_{[i]}) \geq m \cdot 2^n$.*[4]

We can now prove that some specific formulas equivalent to F_n (see Definition 6) have a size exponential in n.

▶ **Proposition 17.** *Let $n > 0$ be a natural number. Let Ts be a list of formulas. Let $G = \bigvee_{i=0}^{|Ts|} Ts_{[i]}$. If all formulas in Ts are conjunctive and satisfiable, and if F_n is equivalent to G, then $\mathsf{sizef}\ G \geq n \cdot 2^n$.*

Proof. We proceed with five steps: **(1)** First, we show that every conjunctive formula $T \in \mathsf{set}\, Ts$ contains for all $i \in \{1, \ldots, n\}$ either the variable $x_{i,0}$ or $x_{i,1}$ positively. We prove this step by negation, where we have to derive false for two cases: Either the variables $x_{i,0}$ and $x_{i,1}$ are both positively contained in T or they are both not positively contained. In both cases, we proceed by defining a valuation that satisfies G but not F_n. For instance, in the second case, we find a valuation that satisfies T and change the assignment of the variables $x_{i,0}$ and $x_{i,1}$ to false. We then show using Lemmas 8, 14, and 15 that this modified valuation satisfies T (and thus G) but does not satisfy F_n. This means that F_n is not equivalent to G, which contradicts the proposition's assumption. **(2)** From step 1, we derive that every $T \in \mathsf{set}\, Ts$ has $\mathsf{sizef}\ T \geq n$. **(3)** Furthermore, we prove that for every Boolean list $\epsilon = [\epsilon_1, \ldots, \epsilon_n]$ there exists a $T \in \mathsf{set}\, Ts$ that contains the corresponding variables $x_{1,\epsilon_1}, \ldots, x_{n,\epsilon_n}$ positively. If this is not the case for a specific list ϵ, we show using step 1 that every T must contain one of the variables $x_{1,\neg\epsilon_1}, \ldots, x_{n,\neg\epsilon_n}$ positively. We then define a valuation that maps all variables x_{i,ϵ_i} to true and $x_{i,\neg\epsilon_i}$ to false for $i \in \{1, \ldots, n\}$. We show using Lemma 15 that this valuation satisfies F_n but not G. This means that F_n is not equivalent to G, which contradicts the proposition's assumption. **(4)** We then prove that the number of Boolean lists of length n is $2^n \leq |\mathsf{set}\, Ts| \leq |Ts|$ using steps 1 and 3. **(5)** Finally, we show that $\mathsf{sizef}\ G \geq n \cdot 2^n$ by combining the results from steps 2 and 4 to discharge the assumptions of Lemma 16. ◀

[4] The function $\mathsf{set} :: {'a}\ list \Rightarrow {'a}\ set$ converts a list into a set.

Proposition 17 assumes the existence of two equivalent formulas F_n (in conjunctive normal form) and $\bigvee_{i=0}^{|Ts|} Ts_{[i]}$ (in disjunctive normal form) and proves a lower bound on the size of the later. We now prove that such equivalent formulas do exist and that the exponential blowup holds for all of them.

▶ **Theorem 18** (Exponential Blowup from CNF to DNF)**.** *Let n be a natural number. There exists a formula ϕ in conjunctive normal form with size $|\phi| = 10 \cdot n + 2$ such that every equivalent formula ψ in disjunctive normal form has size $|\psi| \geq n \cdot 2^n$.*

Proof. We start by a case analysis on n. The interesting case is when $n > 0$. We define $\phi = F_n$ (see Definition 6); it is easy to show that F_n is in conjunctive normal form (using Lemma 7) and that $|\phi| = 10 \cdot n + 2$ (using Lemma 12). To prove the last property, we assume that there is an equivalent formula ψ in disjunctive normal form. We first obtain a list Ts (by filtering $\mathsf{undisj}\ \psi$) of all satisfiable terms in ψ such that all formulas in Ts are conjunctive and satisfiable, $\mathsf{sizef}\ \psi \geq \mathsf{sizef}\ (\bigvee_{i=0}^{|Ts|} Ts_{[i]})$ (using Lemma 13), and $\bigvee_{i=0}^{|Ts|} Ts_{[i]}$ is equivalent to ψ. From this last equivalence, we show that $\bigvee_{i=0}^{|Ts|} Ts_{[i]}$ is equivalent to F_n. This allows us to prove that $\mathsf{sizef}\ (\bigvee_{i=0}^{|Ts|} Ts_{[i]}) \geq n \cdot 2^n$ using Proposition 17. We can prove that $\mathsf{sizef}\ \psi \geq n \cdot 2^n$ by transitivity and, finally, that $|\psi| \geq n \cdot 2^n$ using Lemma 11. ◀

We claim that Theorem 18 is a formalization of the paper proof of Theorem 2.

5 DNF to CNF (MDS)

Our formalization splits the paper proof of Theorem 3 again in a low-level proposition, which assumes the existence of formulas with specific shapes and properties, and a theorem, which proves that such formulas can always be obtained. We start by defining a function to produce the dual of a formula and proving a few basic properties.

▶ **Definition 19.** The function $\mathsf{dualize} :: {}'a\ formula \Rightarrow {}'a\ formula$ converts a formula to its dual form by replacing each literal with its complementary: an atom A becomes $\neg A$, falsehood $\bot$ becomes $\neg\bot$, a negation $\neg\phi$ becomes ϕ, a disjunction $\phi \vee \psi$ becomes $\mathsf{dualize}\ \phi \wedge \mathsf{dualize}\ \psi$, a conjunction $\phi \wedge \psi$ becomes $\mathsf{dualize}\ \phi \vee \mathsf{dualize}\ \psi$, and an implication $\phi \to \psi$ becomes $\phi \wedge \mathsf{dualize}\ \psi$.

▶ Remark. In contrast to the definition of dual formulas in Section 2, we specify the $\mathsf{dualize}$ function for arbitrary formulas to avoid the recurring assumption that formulas are in negation normal form.

The semantics of the dual of a formula is the exact opposite of that of the original formula.

▶ **Lemma 20.** *Let ϕ be a formula. The formula $\mathsf{dualize}\ \phi$ is equivalent to $\neg\phi$.*

Because the sizef function ignores negations, the $\mathsf{dualize}$ function preserves the size of the original formula.

▶ **Lemma 21.** *Let ϕ be a formula.* $\mathsf{sizef}\ (\mathsf{dualize}\ \phi) = \mathsf{sizef}\ \phi$.

Next, we prove some sufficient condition for a dual formula to be satisfiable.

▶ **Lemma 22.** *Let ϕ be a formula. If ϕ is disjunctive and not tautological, then $\mathsf{dualize}\ \phi$ is satisfiable.*

Next, we prove that the dual of some specific formulas in disjunctive normal form are in conjunctive normal form.

▶ **Lemma 23.** *Let Cs be a list of formulas. If all formulas in Cs are disjunctive and not tautological, then there exists a list Ts such that all formulas in Ts are conjunctive and satisfiable, and* $\mathsf{dualize}\ (\bigwedge_{i=0}^{|Cs|} Cs_{[i]}) = \bigvee_{i=0}^{|Ts|} Ts_{[i]}$.

Proof. We prove this by structural induction over the list Cs using Lemma 22. The idea is to map each element of Cs to its dual form. ◀

We can now prove that some specific formulas equivalent to the dual of F_n (see Definition 6) have a size exponential in n.

▶ **Proposition 24.** *Let $n > 0$ be a natural number. Let Cs be a list of formulas. Let $G = \bigwedge_{i=0}^{|Cs|} Cs_{[i]}$. If all formulas in Cs are disjunctive and not tautological, and* $\mathsf{dualize}\ F_n$ *is equivalent to G, then* $\mathsf{sizef}\ G \geq n \cdot 2^n$.

Proof. We prove this by contradiction using Proposition 17. We assume $\mathsf{sizef}\ G < n \cdot 2^n$ and from that derive false in five steps. **(1)** First, we know from Lemma 21 that $\mathsf{sizef}\ (\mathsf{dualize}\, G) = \mathsf{sizef}\ G$, thus $\mathsf{sizef}\ (\mathsf{dualize}\ G) < n \cdot 2^n$. **(2)** Moreover, we show that $\mathsf{dualize}\ G$ is equivalent to F_n using, among others, Lemma 20. **(3)** Also, we prove using Lemma 23 that there exists a list Ts, where every formula contained in Ts is both conjunctive and satisfiable, such that $\mathsf{dualize}\ G$ is syntactically equal to $\bigvee_{i=0}^{|Ts|} Ts_{[i]}$. **(4)** From steps 2 and 3, it follows that Proposition 17 holds for F_n and $\mathsf{dualize}\ G$. According to Proposition 17, it therefore must be the case that $\mathsf{sizef}\ (\mathsf{dualize}\ G) \geq n \cdot 2^n$. **(5)** The result of step 1 contradicts the result of step 4, completing the proof. ◀

Proposition 24 assumes the existence of two equivalent formulas $\mathsf{dualize}\ F_n$ (in disjunctive normal form) and $\bigwedge_{i=0}^{|Cs|} Cs_{[i]}$ (in conjunctive normal form) and proves a lower bound on the size of the later. We now prove that such equivalent formulas do exist and that the exponential blowup holds for all of them.

▶ **Theorem 25** (Exponential Blowup from DNF to CNF)**.** *Let n be a natural number. There exists a formula ϕ in disjunctive normal form with size $|\phi| = 10 \cdot n + 1$ such that every equivalent formula ψ in conjunctive normal form has size $|\psi| \geq n \cdot 2^n$.*

Proof. The proof is analogous to the proof of Theorem 18. ◀

We claim that Theorem 25 is a formalization of the paper proof of Theorem 3.

6 Discussion

Whereas most ideas of the paper proof can be directly used for the formalization, some intuitive mathematical proof steps required significant formalization effort. This was most pronounced in the proof steps 2 and 4 of Proposition 17, where we derived statements about the size of a formula from its properties. To prove step 4, we use a general lemma (already included in Isabelle/HOL) about the cardinality of finite sets.

▶ **Lemma 26.** *Let X and Y be sets. Let f be a unary function. If Y is finite, and if $\{f\ x \mid x \in X\} \subseteq Y$, and if f is injective on X, then the cardinality of X is less than or equal to the cardinality of Y.*

We use Lemma 26 to prove that the number of Boolean lists of length n is less than or equal to the cardinality of $\mathsf{set}\ Ts$. We instantiate X with the set of all Boolean lists of length n, Y with $\mathsf{set}\ Ts$, and f with a function that takes a Boolean list and returns some conjunctive formula $T \in \mathsf{set}\, Ts$ for which the predicate of step 3 holds. Finding an appropriate function for f and discharging Lemma 26's assumptions was surprisingly challenging: The first assumption was the most intricate and its proof uses both steps 1 and 3, the second assumption follows from step 3, the third assumption easily holds.

To prove step 2, we introduced a new lemma, inspired by Lemma 26, that uses a binary function that is right total w.r.t. the target set.

▶ **Lemma 27.** *Let X be a set of elements of type $'a$ and Y be a set of elements of type $'c$. Let $f :: 'a \Rightarrow 'b \Rightarrow 'c$ be a binary function. If X and Y are finite sets, and if for all $x \in X$ there exists a y such that $f\ x\ y \in Y$, and if the function f is injective on all those pairs of x and y, then the cardinality of X is less than or equal to the cardinality of Y.*

We use Lemma 27 to prove that every $T \in \mathsf{set}\, Ts$ contains at least n atoms and thus that $\mathsf{sizef}\ T \geq n$. To do so, we instantiate X with $\{1, \ldots, n\}$, Y with the set of atoms of T, and the f with the constructor for variables. Finding the correct statement for Lemma 27 was challenging but, once found, it was easy to discharge the assumptions using step 1.

Moreover, we chose to modify the functions BigAnd and BigOr from Michaelis and Nipkow [4] and to define our own versions in Definition 4, where the base cases are handled differently. Our functions avoid appending $\neg\bot$ and $\bot$ to the produced formula when the provided list is nonempty. This allowed us to prove the size equalities in Lemma 13 instead of having the left-hand side of the lemmas be slightly smaller than the right-hand side. Using the original functions would have been possible, but it would have resulted in Theorems 18 and 25 stating that the size $|\psi| + 2 \geq n \cdot 2^n$ instead of $|\psi| \geq n \cdot 2^n$. To keep the statements as close as possible to their mathematical counterparts, we therefore chose to introduce the functions BigAnd' and BigOr'.

Finally we introduced the function sizef to be able to prove the equality of Lemma 21. Without sizef, we could only prove the inequality $|\mathsf{dualize}\ \phi| \leq 2 \cdot |\phi|$ because dualize adds and removes negations depending on the formula. It would be possible to restate and prove the statement of Proposition 17 in terms of $|\cdot|$. However, if we tried to restate and prove the statement of Proposition 24 in terms of $|\cdot|$, we would prove $|\mathsf{dualize}\ G| < 2 \cdot n \cdot 2^n$ in step 1 and $|\mathsf{dualize}\ G| \geq n \cdot 2^n$ in step 4, and we would not be able to derive a contradiction in step 5.

7 Conclusion

We formalized in Isabelle/HOL the well-known folklore result about propositional logic that transforming a formula in disjunctive or conjunctive normal form can lead to an exponential blowup of the formula size. The proof is split into a low-level proposition and a theorem for conversion to disjunctive normal form and analogously for conversion to conjunctive normal form. We also reported of some challenges we faced we encountered when proving statements about the size of formula.

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